

AMJAD ALOLAYAN

M.Sc. Sustainable Urban Development · GIS & Geospatial Analysis

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SUMMARY

Master's student in Sustainable Urban Development at Technical University Darmstadt with international experience in Japan, Vietnam, and Germany. Strong interest in GIS analysis and sustainable infrastructure systems, with growing expertise in Python and ArcGIS Pro for data-driven urban sustainability applications.

GIS PROJECTS

[\(Portfolio →\)](#)

Multi-Criteria Solar Park Suitability Analysis – Wetteraukreis & Vogelsbergkreis, Hesse

2026

Personal Project · ArcGIS Pro, Python (GeoPandas, Shapely, STRtree, Folium)

- Built a multi-criteria GIS suitability model for solar parks across two Hessian districts, applying six hard exclusions and scoring 197 candidates on five weighted criteria via zonal statistics and STRtree analysis in Python; produced ArcGIS Pro layouts and a live Folium/Leaflet web map published on GitHub Pages.

Rooftop Solar Potential Analysis – Breuberg, Hesse

2026

Academic coursework, independently rebuilt & corrected · ArcGIS Pro, PVGIS, OpenStreetMap

- Rebuilt a graded university solar-energy assignment in ArcGIS Pro using PVGIS 5.3 (PVGIS-SARAH3) irradiation data and OpenStreetMap building footprints; found that large industrial rooftops account for a disproportionate share of technical potential.

WORK EXPERIENCE

Compliance Officer, Gulf International Bank

Apr 2022 – Sep 2023

Al Khobar, Saudi Arabia

- Monitored client transactions via CRM and Oracle FCC systems; conducted policy and risk analysis supporting regulatory compliance.
- Conducted internal audits/reviews to identify operational gaps and implement corrective action plans.

Intern, National Center for Environmental Compliance (NCEC)

May – Oct 2021

- Riyadh, Saudi Arabia: supported environmental compliance and inspection projects, monitoring facility adherence to environmental regulations through data collection, policy documentation, and preparation of governmental reports.

Research Intern, JACES (Institute for Global Environmental Strategies)

Jul – Sep 2017

- Tokyo, Japan: conducted policy research and contributed to recommendations under the SDGs/Sustainable Consumption and Production (SCP) Program, including literature review, data analysis, and English–Japanese translation of policy documents.

EDUCATION

M.Sc. Sustainable Urban Development – TU Darmstadt

Oct 2023 – Present (exp. 2027)

Darmstadt, Germany · Focus on sustainable urban/infrastructure development, climate mitigation, green building design, GIS, and Python-based environmental data analysis.

Joint Master's Program – Vietnamese-German University

Oct 2024 – Mar 2025

Vietnam · Sustainability, circular economy, decarbonisation strategies; applied GEG-based building energy calculations and cost-benefit analysis.

B.A. Community and Regional Policy Studies – Ritsumeikan University

Sep 2016 – Mar 2021

Osaka, Japan · Fully funded scholarship · Public policy, sustainability, and urban governance.

- Bachelor's thesis: "Adaptability of reverse-vending machines for sustainable waste management in Saudi Arabia," policy analysis and international case studies.

TECHNICAL SKILLS

GIS & Mapping: ArcGIS Pro (spatial analyst, zonal statistics, model building, cartographic layouts), QGIS, kernel density & multi-criteria suitability modelling

Programming: Python (GeoPandas, Shapely, STRtree, Folium, pandas, matplotlib), spatial data automation

Data, Tools & Workflows: Excel, SPSS, openLCA, OpenStreetMap/Overpass API, PVGIS, Git/GitHub, web-map publishing (Leaflet)

LANGUAGES

Arabic: Native · English: C1 · Japanese: C1 · German: A2–B1 (basic working proficiency)

CERTIFICATES

ArcGIS API for Python Fundamentals (Esri, 2026) · ArcGIS Pro Scripting & Spatial Analysis Automation (Esri, 2026)